



RAIL-BLOC — Complete User Walkthrough (Jab Website Ready Hogi)

Poora flow ekdum start se end tak. Isko padh ke frontend design kar sakte ho — har page, har button, har click ka result.

① Sabse Pehle: Login Page (/login)

Kya dikhega: Dark navy (#0B111E) background, center mein white card — "RAIL-BLOC Atlas Console" title, Username box, Password box, "Sign in" button. Neeche chhota text: "SIH26027 · all data is SIMULATED".

Buttons:

- **Sign in** → POST `/api/v1/auth/login` → success pe `/dashboard` redirect, token clipboard/localStorage mein save
- Login fail → red error box: "invalid credentials"

Role ke hisaab se focus: 7 demo users hain:

User	Kya kar sakta hai
<code>srdm_dli</code>	Approve plans
<code>drm_dli</code>	Authorize plans (dusra user ho!)
<code>controller_dli</code>	Emergency trigger + Controller ack
<code>engineer_dli</code>	Manual upload + fitness certify
<code>sm_dli</code>	S&T signal acknowledgment
<code>auditor</code>	Ledger verify
<code>admin</code>	Solve trigger, everything

2 Home: Dashboard (/dashboard)

Kya dikhega: Top mein "Divisional Overview" heading. 4 bade KPI cards:

Card	Kya dikhata hai
Active blocks	🔴 red number = TRANSMITTED_COA + ACTIVE_GRANTED plans count
Predicted pax delay	Amber number (min) + "model estimate (B1-relative, simulated data)" label
Predicted freight detention	Blue number (min) + model estimate label
Escalated overdue	🔴 X number = ESCALATED_OVERDUE demands (FSM-002 cap exhausted)

Neeche 2 sections:

- **Overdue / Escalated demands table** — jinki scheduling 3 attempts fail, human review chahiye. Red/amber rows with urgency score.
- **Machine fleet utilization** — kaunsi machine kitne jobs + travel minutes.

Neeche Plan lifecycle distribution — saare status badges (SENTINEL_PASSED, APPROVED_SR_DOM, PROVISIONAL etc.) with counts.

Auto-refresh: Har 15 sec data refresh hota hai (background polling).

Sidebar navigation (left side, hamesha visible):

Dashboard	Corridor Map	String Chart	26-Week Calendar	Weekly Planner	Approvals	Disrupti
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Role ke hisaab se kuch items chhup jaate hain (sirf APPROVALS roles ko, sirf AUDITOR ko Audit Ledger, etc.)

3 Corridor Map (/corridor-map)

Kya dikhega: Full-width MapLibre map (Google Maps jaisa dark theme). NDLS→CNB corridor ki track lines visible.

Layers:

- ● Neele line = track centerline (normal)
- ● Rati red = **blocked section** (live block)
- ● Dashed cyan = **OHE feeding boundaries** (power isolation)
- 🚂 Moving train markers (every 10s update — RTIS mock, timetable se interpolate)

Interactions:

- **Hover section** → popup: section code, KM range, GMT index, defect count, *"SIMULATED health indices"*
- **Click section** → popup larger
- **Zoom/pan** mouse se

Buttons:

- Zoom controls (+/-) top-right
- (No action buttons — purely view + LIVE)

Important: Hamesha bottom-right **"SIMULATED DATA"** watermark, aur agar SSE stream drop ho toh top pe **"STALE DATA — actions disabled"** red banner.

4 String Chart (/string-chart)

Kya dikhega: Full-width HTML5 Canvas chart. X-axis = distance (0–250 km), Y-axis = time (00:00–23:59 = 24h window perfect).

Elements:

- ● Purple diagonal lines = Vande Bharat/Rajdhani (priority 1-2)
- ● Blue lines = Mail/Express (priority 3-4)

-  Amber lines = Passenger + Freight
-  Semi-transparent rectangles = **block windows** (time × km position)
-  Diagonal stripes wale boxes = **shadow bundles** (multi-department co-allocated)

Buttons/Controls:

- ◀ **prev day / today / next day** ▶ — day navigation
- **Scroll** = zoom (distance axis)
- **Ctrl+Scroll** = zoom (time axis)
- **Drag** = pan
- **Hover** = tooltip: jog KM + section + GMT + defects + scheduled block info

5 26-Week Calendar (/planner/26-week)

Kya dikhega: Grid table. Rows = 12 sections (NDLS-GZB-UP, GZB-ALJN-DN, etc.), Columns = 26 weeks (W2026-08-26 se...).

Cells mein: Status badge chips (SENTINEL_PASSED, AUTHORIZED_DRM, PROVISIONAL etc.) + revision number.

Minimal buttons:

- Sirf view — no actions. Strategic 26-week horizon plans database se aate hain. Agar empty ho: message *"No strategic plans yet — trigger a STRATEGIC_26W solve from the weekly planner."*

6 Weekly Planner (/planner/weekly) — THE BIG ONE

Kya dikhega: 2-column layout — left: plans list, right: selected plan detail.

Left side (plans list):

- ▶ **"Trigger weekly solve"** button (top-right) — POST `/optimize/solve` → 202 task id → 8-20 sec baad list refresh
 - Click hote hi: "Solve queued: task_id..." message → background polling → naye plans aate hain
- Plan rows: section code, time range, `rev1` badge, **"est. pax Xm · frt Ym (model estimate)"**

Right side (Modify Parameters):

- Selected plan ka detail: status badge, content_hash (chhota), **New start / New end** datetime inputs
- 📝 **"Revise → re-enter Sentinel chain"** button:
 - POST `/plans/{id}/revise` → naya revision banta hai (rev2) — purana plan SUPERSEDED, naya DRAFT + `sentinel_verified: false`
 - Message: *"Revision created: rev 2 (...) — new revision re-enters the Sentinel chain"*
- Neeche chhota text: *"FR-026: any mutation after SENTINEL_PASSED creates revision+1 at DRAFT, clears sentinel_verified and restarts the approval chain (SAFE-002)."*
- Status badge update hogi: `DRAFT` (spinning/pulsing)

Behind the scenes (solve pipeline):

POST /optimize/solve

- Redis lock (division+horizon) [agar lock busy: "a solve already running" 409]
- solver_runs table mein QUEUED row
- Celery worker picks it up
- Weather fail-closed check (stale IMD → outdoor work deferred)
- Demands (SUBMITTED/NORMALIZED) load karta hai
- Optional ML urgency refresh (PyTorch)
- B1 greedy warm-start hint
- OR-Tools CP-SAT solve (default 35s budget)
- Sentinel 10-check validation
- Sab pass → plans persist (SENTINEL_PASSED ya DRAFT agar GSR-2 pending)
- machine_rosters persist
- Ledger event + SSE publish
- solver_runs COMPLETED

7 Approvals (/approvals) — SAFETY-CRITICAL SCREEN

Kya dikhega: 2-column — left: queue, right: **Action Preview Card**.

Left (queue): Plans `SENTINEL_PASSED` (Sr. DOM ke liye) aur `APPROVED_SR_DOM` (DRM ke liye).
Click select karo.

Right — Action Preview Card (ye [Design.md](#) ka hero component hai):

Action Preview	[STATE BADGE]
WHAT: NDLS-GZB-UP	
01:30 - 05:00 · 210 mins · rev1	
hash: 4f8ab2... (chhota)	
WHY: DTT_TAMPING $\Pi \approx 0.812$	
⌚ TEL-001 freshness: ingested	
2026-08-26 01:00Z (2h ago) [green]	
SHADOW CLUSTER: 2 co-allocated works	
(Track Tamping + OHE Cantilever)	
IMPACT ANALYSIS:	
predicted pax 55m · frt 120m	
[model estimate (B1-relative, simulated data)] – italic label	
SAFETY VERIFICATION: 10/10 CHECKS	
✓ 1. G&SR-1 Absolute Block	
✓ 2. G&SR-2 Interlocking Precedence	
✓ 3. G&SR-3 Fail-Closed Consistency	
✓ 4. G&SR-4 Power Isolation	
✓ 5. G&SR-5 Headway Margin	
✓ 6. MILP-C1 Section Exclusion	
✓ 7. MILP-C2 Maintenance Enclosure	
✓ 8. MILP-C3 Shadow Bundling	
✓ 9. MILP-C4 Non-Fragmented	
✓ 10. MILP-C5 Machine Conservation	
REVISION INTEGRITY:	
local 4f8ab2... · server 4f8ab2...	
decided_by [name]	
authorized_by: – pending–	
[✓ Approve & Digitally Sign]	
[X Reject Plan]	

Buttons & kya hota hai:

- ✓ **Approve & Digitally Sign** (Sr. DOM):
 - POST `/approvals/decide` {plan_id, decision: APPROVE, signature, idempotency_key}
 - Server: hash re-verify → status SENTINEL_PASSED → **APPROVED_SR_DOM**, `decided_by` record
 - Response: *"APPROVED_SR_DOM · ledger 0x4f8ab2..."*
 - **Hash mismatch** → red banner: *"Plan changed — reload to review latest revision. Approve disabled while the locally-held hash is stale."* — button disabled
 - **Same actor self-authorize (DRM)** → red: *"distinct-approver violation"*
- **(DRM ke liye button label change)** → [ **Authorize & Seal**] — DRM step, different label!
 - POST `decide` → status **AUTHORIZED_DRM**, `authorized_by` record
 - Footer shows: `decided_by: srdom_dli · authorized_by: drm_dli`
- ✗ **Reject Plan** → status CANCELLED, demands CANCELLED
- Agar plan DRAFT (S&T pending acks) → yellow info box: *"G&SR-2 pending acknowledgments"* + **"Acknowledge as Station Master"** / **"Acknowledge as Controller"** buttons (sirf STATION_MASTER/CONTROLLER roles ko)

Special case — S&T plan (G&SR-2):

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DRAFT (signals pending)
→ SM acknowledge → controller acknowledge
→ plan → SENTINEL_PASSED (hash sealed)
→ phir approve → authorize flow
```

⑧ Disruptions (`/disruptions`) — EMERGENCY CONSOLE

Kya dikhega: Left: incident log form + list. Right: semantic notes card.

Button flow:

1. ✂ **Preview blast radius** button:

- Section ID select (12 sections)
- Type select (TRACK_FRACTURE, OHE_BREAKDOWN, SIGNAL_FAILURE, OTHER)
- Duration input (15-1440 min)
- Click → **Blast Radius Modal** khulta hai:

⚠ Emergency Confirmation Modal – blast radius

- Trains currently held: 3
 - Plans that will be superseded: 2
 - Affected sections (incl. adjacent via feeding map): 4
- [] I acknowledge the blast radius above (API-001 modal).
Advisory revocations will be issued by the Emergency Service;
the re-plan is PROVISIONAL until I acknowledge it.
- [🔥] Fire emergency breakdown

- Checkbox ke bina "Fire" button disabled — **pehle confirmation chahiye** (API-001)

2. 🔥 **Fire emergency breakdown:**

- POST `/emergency/breakdown` → 201
- System kya karta hai (behind the scenes):

1. Section pe active/planned block check (pass/fail)
2. Incident create (adjacent sections check → coalesce)
3. Advisory revocation → TRANSMITTED_COA/ACTIVE_GRANTED plans → SUPERSEDED_
4. Displaced demands → SUBMITTED
5. Corridor-scoped CP-SAT re-plan (≤35s budget)
6. Sentinel structural checks (synchronous, sub-second)
7. Plan persist → PROVISIONAL (incident bound) + ledger + SSE
8. Response + measured wall time

- Response message: *"✓ PROVISIONAL plan 8f2ab1... created in 12.3s (NFR-002 budget incl. synchronous structural re-check). Coalesced: no. Superseded: 2 plan(s). Awaiting Controller acknowledgment."*

3. **PROVISIONAL plan item** → purple badge + **"Acknowledge as Controller"** button:

- POST `/emergency/incidents/{id}/acknowledge`

- → incident ack → outbox transmit allowed → COA bridge ack → TRANSMITTED_COA
- Controller ke ack ke bina plan ka kuch nahi hota (deadlock gate — by design)

List mein incidents: Section, type, est duration, coalesced tag (📅), provisional plan id, controller ack status.

9 Audit Ledger (/audit-ledger)

Kya dikhega: Full-width table (sirf AUDITOR/ADMIN).

seq	event	actor	prev_seq	prev_hash→hash	payload
12	PLAN_SENTINEL_PASSED	worker	11	4f8ab...→0x...	{"plan_id":...}

Buttons:

- 🔒 Verify chain (REPEATABLE READ):
 - GET /ledger/verify → result card:

✓ tamper-EVIDENT chain intact – verified 54/54

Method: full re-hash from sequence 1 under REPEATABLE READ snapshot isolation
 - Broken chain → red "X CHAIN BROKEN"
- Search input (event_type filter)
- Infinite scroll/pagination

Neeche chhota text: "tamper-evident, not tamper-proof — [Rules.md §3](#)"

🔄 COMPLETE E2E DEMO FLOW (Demo script ke liye — 5 min)

Time	Kya karte ho	Kya hota hai system mein
0:00	Login <code>srdom_dli</code>	JWT mint hua
0:20	Dashboard	KPIs + escalated table
0:40	Weekly Planner → Trigger solve	CP-SAT solve, 8-20s
1:00	Plans list refresh	SENTINEL_PASSED plans dikhte hain
1:15	Approvals → plan select	Action Preview Card: 10/10 checks
1:30	Approve (Sr. DOM)	APPROVED_SR_DOM + decided_by
1:45	Login <code>drm_dli</code> → Authorize & Seal	AUTHORIZED_DRM + authorized_by (distinct!)
2:00	Login <code>controller_dli</code> → Plans → plan info	Plan AUTHORIZED_DRM
2:15	Transmit button	T-2h structural re-check → outbox → COA ack → TRANSMITTED_COA
2:30	Activate → Complete-fitness → Archive	ACTIVE_GRANTED → COMPLETED_FITNESS → ARCHIVED_SEALED
3:00	Disruptions → Preview → Fire breakdown	Blast radius modal → PROVISIONAL plan
3:20	Acknowledge as Controller	PROVISIONAL → transmit → TRANSMITTED_COA
3:40	Login <code>auditor</code> → Verify chain	54/54 unbroken

Time	Kya karte ho	Kya hota hai system mein
4:00	Modify-after-verify attempt (pehle wala plan revise karo)	rev2 → DRAFT; approve kiro → 409 HASH_MISMATCH
4:30	SIMULATED DATA watermark dikhao	Honesty label
5:00	Summary: "ML estimates; CP-SAT decides; Sentinel verifies; humans authorize; COA executes."	Wrap-up

🛡️ HARR BHI KYA HAIN — STATUS BADGES (har jagah same)

Badge	Color	Matlab
DRAFT	Gray ○	Abhi banaya, Sentinel pending
SENTINEL_PASSED	Green 🛡️	10/10 checks pass, approval ready
APPROVED_SR_DOM	Blue ✔️	Sr. DOM ne decide kiya
AUTHORIZED_DRM	Amber 🔒	DRM ne authorize kiya
TRANSMITTED_COA	Blue 📡	COA ko mila (ack hua)
ACTIVE_GRANTED	Red ⚡	Line isolated, work chalu
COMPLETED_FITNESS	Green ✔️	Khuli line
ARCHIVED_SEALED	Gray 📦	History sealed
SUPERSEDED	Gray ↗️	Revision replace hua
SUPERSEDED_EMERGENCY	Amber ⚠️	Emergency ne replace kiya
CANCELLED	Gray ✖️	Reject/cancel
FAILED_ESCALATE	Red ✖️	3 attempts fail — human review

Badge	Color	Matlab
PROVISIONAL	Purple ♦	Emergency plan, Controller ack pending

Har badge automatically 3 cheezein dikhata hai: icon + text + color (WCAG 1.4.1 colorblind-safe).

🚀 IMPORTANT DESIGN POINTS

1. **STALE DATA overlay:** SSE stream drop hote hi (Redis down/crash) → **sab action buttons DISABLED** + persistent top banner. Controller kabhi bhi stale data pe action nahi kar sakta.
2. **SIMULATED DATA watermark:** Hamesha bottom-right, non-dismissible (har synthetic layer pe).
3. **Model estimates label:** Har delay figure ke saath "*model estimate (B1-relative, simulated data)*" — fake fact kabhi nahi.
4. **Hash mismatch banner:** Approve button disabled agar local hash $\neq$ server hash.
5. **Distinct approver:** DRM button label ALAG ("Authorize & Seal") + footer mein decided_by/authorized_by display.
6. **Emergency modal:** BLAST RADIUS dikhana mandatory — confirmation checkbox ke bina fire nahi ho sakta.

🏗️ FRONTEND ARCHITECTURE (design ke liye)

- **Pages (9):** Login, Dashboard, CorridorMap, StringChart, Planner26Week, PlannerWeekly, Approvals, Disruptions, AuditLedger
- **Shared components:** NavDrawer (role-filtered), StatusBadge, Card, ActionButton (stale pe disabled), PreviewCard, StaleOverlay, SimulatedWatermark
- **Live events (SSE):** BLOCK_ACTIVATED, PLAN_REVISED, SIGNAL_ACK, BLOCK_TRANSMITTED, PROVISIONAL_PLAN_CREATED, SOLVE_COMPLETED, SOLVE_FAILED — inko live badges/markers update karne ke liye use kar sakte ho

- **Auth:** localStorage token, JWT claims role/division — nav + permissions isi se
 - **Strict TypeScript** — `tsc --noEmit` build gate
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Ab jo bhi design karna hain — login card, dashboard grid, map, string chart, 10-check card, emergency modal, ledger table, badges — sab ka exact behavior upar hai. Isi hisaab se polish kar do, aur frontend-ready. Koi aur cheez chahiye toh bolo! 🙏